

Harrison Felstead

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EDUCATION

Monash University

Melbourne, VIC

Bachelor of Engineering (Hons) (Mechatronics) / Bachelor of Science (Computer Science) 2022 – Present

St Augustine's Marist College

Cairns, QLD

Queensland Certificate of Education | ATAR: 97.80

2018 – 2021

TECHNICAL SKILLS

Programming	Python, C++, C, Assembly, MATLAB
Frameworks & Tools	PyTorch, Jupyter, Git, Linux
Hardware / CAD	CAD (SolidWorks), Arduino, Raspberry Pi, STM32
Software	Microsoft Office Suite, Agile/Scrum project management tools
Languages	English (Native), German (A2)

WORK EXPERIENCE

OSHC Educator

Summer 2025–2026

Holy Cross Primary School – Cairns

- Supervised and guided primary school students in structured activities, developing communication and patience skills applicable to client-facing engineering roles.

ENGINEERING & TECHNICAL PROJECTS

Research & Development Team Member

April 2026 – Present

Monash BrewLab

Monash University

- Designed and tested multiple fermentation combinations using food chemistry principles, applying a scientific methodology to evaluate, iterate, and optimise outcomes.
- Applied data analysis techniques to assess fermentation yields and flavour profiles, translating experimental results into actionable process improvements.
- Collaborated within a multidisciplinary team to document findings and plan future R&D directions.

Final Year Project – Custom Orthopaedic sole creation using Unsupervised Learning

- Developed an unsupervised Machine Learning pipeline using K-means clustering to segment plantar pressure maps into superpixels representing regions of similar pressure magnitude and pressure change dynamics.
- Generated custom 3D-printable sole geometries from clustering outputs, translating biomechanical data into manufacturable designs tailored to individual foot profiles.
- Project spans Machine Learning, Computer Vision, biomechanical analysis, and additive manufacturing forming an end to end from raw data through to physical output.

Mechatronics & Computer Science Coursework Projects

Monash University

- Built practical experience across embedded systems (C/C++), control systems design, robotics, signal processing, and machine learning (PyTorch), with CAD applied across mechanical design and prototyping units.

LEADERSHIP & EXTRACURRICULAR

Vice President

Sep 2024 – Present

German Liechtenstein Austrian Swiss Society (GLASS)

Monash University

- Overseeing committee operations, facilitating decision-making, and supporting strategic planning for the society's annual program.
- Coordinating cross-cultural events and managing a diverse committee, building intercultural communication and organisational skills.

Education Officer

Sep 2024 – Sep 2025

Monash Engineering Students Society (MESS)

Monash University

- Created the "Ed Guide", a first-year resource adopted by the Engineering faculty to help incoming students navigate academic expectations and university life.
- Provided peer mentoring and academic support to first-year Engineering students.

Wellbeing & Equity Officer

Sep 2023 – Sep 2024

Monash Engineering Students Society (MESS)

Monash University

- Developed and launched initiatives to promote mental health awareness within the Engineering faculty.
- Designed and implemented inclusivity and wellbeing programs serving a student body of thousands.

Volunteer – Rosies Friends on the Street

2014 – 2017

Cairns, QLD

- Prepared and distributed food to vulnerable and homeless community members, demonstrating a sustained commitment to community service from a young age.

CERTIFICATIONS

Google Project Management Certificate – Coursera

Dec 2023

- Agile and Scrum framework application; team management program design.

Google IT Support Certificate – Coursera

Dec 2023

- Network protocols, system administration, and network security principles.

Working with Children Check – Card No. 2681764A-01

REFERENCES

Available upon request.